

Day 8: Intro to Python (Cybersecurity Python)

Today, I started the Python section for cybersecurity. I focused on the basics: creating a Python file in Mousepad, writing my first print statements, understanding strings and comments, and then practicing simple math operators.

Creating the first Python file

I created my first Python file using Mousepad:

```
mousepad first.py
```

At the top of the file, I added a shebang line. In my notes I wrote something like "#!/bin/python", but since I'm running Python 3, I learned the clean version is:

```
#!/bin/python3
```

Strings and print()

The first thing I practiced was printing strings. I learned that print() is used to show output in the terminal.

```
print("Hello, World")
```

I also learned that I can use either double quotes or single quotes for strings, and both work the same for simple strings:

```
print("Hi")  
print('Hi')
```

To print multiple lines together, I learned I can use triple quotes:

```
print("""This is a string  
Run multiple lines  
Using triple quotes""")
```

If I want to join text together, I learned I can concatenate (combine) strings using the plus (+) symbol:

```
print("This is " + "awesome")
```

For new lines inside a string, I learned I can use \n. If I want an empty line, I use \n\n:

```
print("\n")
```

Notes and comments

If I want to write notes for myself inside the Python file, I learned I can use # for comments. Python ignores anything after # on that line.

```
# This is my Python note  
# Make notes inside Python
```

Running the script

To run the Python file, I used python3 followed by the file name:

```
python3 first.py
```

```
1 #!/bin/python3  
2  
3 #STRINGS  
4 #print string  
5 print("Hello, world!")  
6 print('hello,world!')  
7 print("""This string runs  
8 multiple lines!""") #triple quote for multi- line  
9 print("This string is "+"awesome!") #We can also concatenate  
10 print('\n') #new line  
11 print('Test that new line out.')  
12
```

Math basics (with print)

After strings, I practiced math. I learned I can do math inside print(), and Python will calculate it and show the result.

Basic operations I practiced:

```
print(50 + 50)
print(50 - 50)
print(50 * 50)
print(50 / 50)
```

I learned that using one slash (/) does division that can return a float (with decimals). Example:

```
print(50 / 6)
```

I also practiced the order of operations. In my notes I called it PMAS, but I learned Python follows the normal math rules (usually written as PEMDAS: Parentheses, Exponents, Multiplication, Division, Addition, Subtraction). Example:

```
print(50 + 50 - 50 * 50 / 50)
```

For exponents, I learned Python uses ** (double star):

```
print(50 ** 2)
```

I practiced modulo (%) too. I learned it gives the remainder:

```
print(50 % 6)
```

I also learned about using // (floor division). It gives the division result without decimals:

```
print(50 // 6)
```

Proof on Medium

<https://medium.com/@mohamd2001momani/day-8-intro-to-python-cybersecurity-python-4bf844341040>

```
14
15 #MATH
16 print(50 + 50) #add
17 print(50 - 50) #subtract
18 print(50 * 50) #multiply
19 print(50 / 50) #divide
20 print(50 + 50 - 50 * 50 / 50 ) #PEMADS
21 print(50 ** 2) #exponents
22 print(50 % 6) #modulo - just the remainder
23 print(50 / 6) #division with repeating digit (or a float)
24 print(50 // 6) #no remainder and no repeating digit
25
```